



ISO/TC 12 "Quantities and units"

Secretariat: SIS

Committee manager: Radhe Sven Mr



Secretariats report 2023

Document type	Related content	Document date	Expected action
Meeting / Secretariat report	Meeting: Norrköping (Sweden) 25 May 2023	2023-05-03	INFO by 2023-05-25



International Organization for Standardization
Organisation internationale de normalisation
Международная организация по стандартизации

ISO/TC 12 Quantities and units

Report of the Secretariat to the meeting of ISO/TC 12 to be held virtual via Zoom, May 25th, 2023.

Year of creation	1947
Secretariat	SIS, Sweden Since 1982, the secretariat is held by SIS.
Chairperson	Dr Torgny Carlsson (until end 2024)
Committee Manager	Mr Sven Radhe, SIS
Secretariat support	Ms Anette Lindor Norén, SIS
ISO Technical Programme Manager	Ms Mercè Ferrés Hernandez
ISO Editorial Manager	Mr Vincenzo Bazzucchi
Scope of ISO/TC 12	Standardization of units and symbols for quantities and units (and mathematical symbols) used within the different fields of science and technology, giving, where necessary, definitions of these quantities and units. Standard conversion factors between the various units.

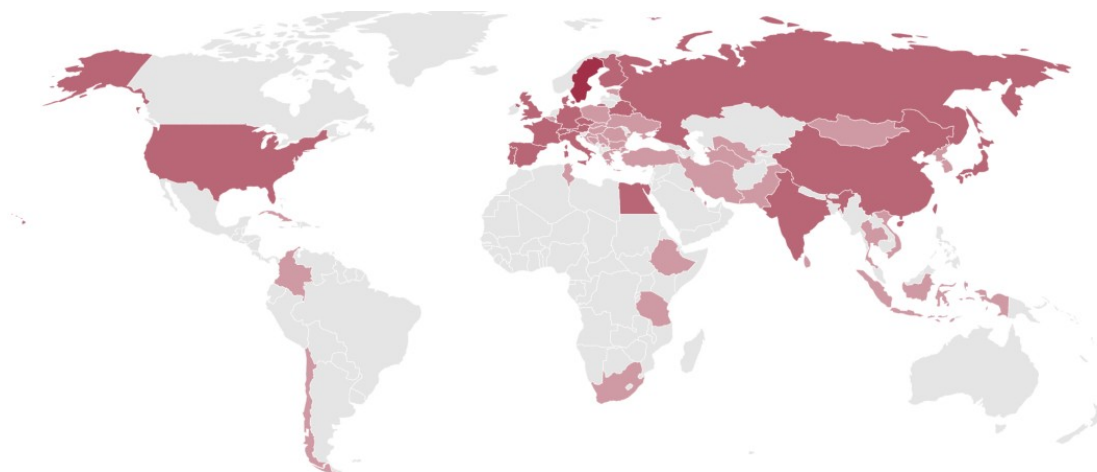
ISO/TC 12 contributes to the following Sustainable Development Goals:

Goal 4: Quality education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

Goal 9: Industry, innovation and infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation



Members 2023-05-02



This map is designed to visually demonstrate the geographic distribution of our Members. The boundaries shown do not imply an official endorsement or acceptance by ISO.

■ Participating Members [20]

■ Observing Members [34]

Austria (ASI)
 Belarus (BELST)
 China (SAC)
 Czech Republic (UNMZ)
 Denmark (DS)
 Egypt (EOS)
 Finland (SFS)
 France (AFNOR)
 Germany (DIN)
 India (BIS)
 Italy (UNI)
 Japan (JISC)
 Kuwait (KOWSMD)
 Portugal (IPQ)
 Russian Federation (GOST R)
 Spain (UNE)
 Sweden (SIS)
 Switzerland (SNV)
 United Kingdom (BSI)
 United States (ANSI)

Belgium (NBN)
 Bosnia and Herzegovina (BAS)
 Bulgaria (BDS)
 Chile (INN)
 Colombia (ICONTEC)
 Croatia (HZN)
 Cuba (NC)
 Estonia (EVS)
 Ethiopia (ESA)
 Greece (NQIS ELOT)
 Hong Kong (ITCHKSAR)
 Hungary (MSZT)
 Indonesia (BSN)
 Iran, Islamic Republic of (ISIRI)
 Korea, Democratic People's Republic of (CSK)
 Korea, Republic of (KATS)
 Mauritius (MSB)
 Mongolia (MASM)
 Pakistan (PSQCA)
 Poland (PKN)
 Qatar (QS)
 Romania (ASRO)
 Serbia (ISS)
 Slovakia (SOSMT)
 South Africa (SABS)
 Sri Lanka (SLSI)
 Tanzania, United Republic of (TBS)
 Thailand (TISI)
 Tunisia (INNORPI)
 Turkey (TSE)
 Turkmenistan (MSST)
 Ukraine (DSTU)
 Uzbekistan (UZSTANDARD)
 Viet Nam (STAMEQ)

NOTE *P-members* (participating members) participate actively in the work, with an obligation to vote on all questions formally submitted for voting within the TC or SC, on new work item proposals, enquiry drafts and final draft International Standards, and to contribute to meetings.

O-members (observers) follow the work as an observer, and therefore receive committee documents and have the right to submit comments and to attend meetings.



New from the last plenary meeting 2022: Moldavia, Republic of CSK is no longer O-member.

Liaison Committees from ISO/TC 12

ISO/TC 12 can access the documents of the committees below:

Reference	Title	Liaison representative
CIE	International Commission on Illumination	Ms. Lindor Norén, Anette Mr. Radhe, Sven
IEC/TC 1	Terminology	Mr. Radhe, Sven
IEC/TC 25	Quantities and units	Dr. Ehrlich, Charles D
ISO/IEC JTC 1/SC 32	Data management and interchange	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 8/SC 6	Navigation and ship operations	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 37	Language and terminology	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 43	Acoustics	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 43/SC 1	Noise	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 47	Chemistry	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 69	Applications of statistical methods	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 85	Nuclear energy, nuclear technologies, and radiological	Ms. Lindor Norén, Anette Mr. Radhe,



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	protection	Sven
ISO/TC 115	Pumps	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 215	Health informatics	Dr. Cullati, Stéphane Prof. Pendrill, Leslie

Liaison Committee to ISO/TC 12

The committees and organisations below can access the documents of ISO/TC 12:

Reference	Title	Liaison representative
IEC/TC 25	Quantities and units	Prof. Mari, Luca
IEC/TC 29	Electroacoustics	Mr Barham, Richard Ms Agesen, Lise Schmidt
ISO/IEC JTC 1/SC 32	Data management and interchange	Ms. Miller, Michaela Ms Rajchel, Lisa Mr Wennblom, Phil
ISO/TC 37	Language and terminology	Mr Cheng, Yonghong Dr Galinski, Christian Mr Wang, Haitao Mr Zhou, Changqing
ISO/TC 43	Acoustics	Mr Kornprobst, Hans-Joachim Ms Kursch, Stephanie Ms Liebing, Daniela
ISO/TC 43/SC 1	Noise	Mr Kornprobst, Hans-Joachim Ms Kursch, Stephanie Ms Liebing, Daniela
ISO/TC 43/SC 3	Underwater acoustics	Mr Ainslie, Michael
ISO/TC 47	Chemistry	Ms Nonaka, Reiko
ISO/TC 69	Applications of statistical methods	Ms Admussen, Jennifer
ISO/TC 85	Nuclear energy, nuclear	Mr Balcaen, Eric



	technologies, and radiological protection	
ISO/TC 215	Health informatics	Dr Geßner, Christof Ms Stanford, Sharon Ms Warner, Diana



Organizations in liaison (Category A and B)
ISO/TC 12 has liaisons with the following organisations:

Acronym	Title	Category	Liaison representative
BIPM	International Bureau of Weights and Measures	A	Dr Milton, Martin Mr Robinson, Stephen Mr Sitton, Robert Mrs de Mirandes, Estefanía Mr Mr Stock, Michael
CIE	International Commission on Illumination	A	Dr Islam, Shahidul Ms Nield, Kathryn
EC - European Commission	European Commission	B	
EURAMET	European Association of National Metrology Institutes	A	Mr Gunn, Robert
IAEA	International Atomic Energy Agency	A	
ICAO	International Civil Aviation Organization	A	
ICRU	International Commission on Radiation Units and Measurements	A	Dr Atwell, Laura
IMO	International Maritime Organization	B	Mr Kishimoto, Isao Mrs Allnutt, Sandra Rita
Infoterm	International Information Centre for Terminology (Infoterm)	A	Dr Galinski, Christian
ITU	International Telecommunication Union	A	Mr Caras, John George Ms Xiaoya Yang, xiaoya (ITU-T SG 17)
IUCr	International Union of	A	Prof Brock, C.P. Mr Ashcroft, Alex



	Crystallography		
IUPAC	International Union of Pure and Applied Chemistry	A	Prof Mills, Ian
IUPAP	International Union of Pure and Applied Physics	A	Dr Karshenboim, Savely Prof Pendrill, Leslie
OASIS	Organization for the Advancement of Structured Information Standards (OASIS)	A	Mr Olken, Frank
OIML	International Organization of Legal Metrology	A	Mr Zechner, Georg
UNECE	United Nations Economic Commission for Europe	A	Mr Henninot, Jean-Pierre Mr Housing, Building and Planning, (UN/ECE)
WCO	World Customs Organization	A	
WHO	World Health Organization	A	Dr Ustun, Bedirhan
WMO	World Meteorological Organization	A	Mr Shi, Peiliang

Category A: Organizations that make an effective contribution to the work of the technical committee or subcommittee for questions dealt with by this technical committee or subcommittee. Such organizations are given access to all relevant documentation and are invited to meetings. They may nominate Experts to participate in a WG (see 1.12.1).

Category B: Organizations that have indicated a wish to be kept informed of the work of the technical committee or subcommittee. Such organizations are given access to reports on the work of a technical committee or subcommittee. Category B is reserved for inter-governmental organizations.



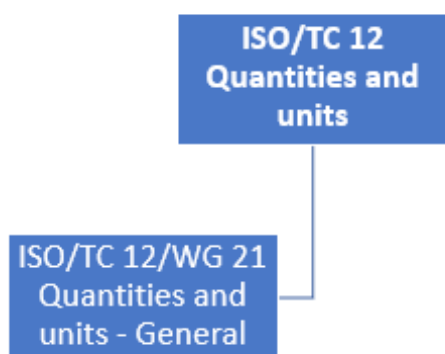
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Meetings, held

1978 Copenhagen, Denmark
 2010 Prague, Czech Republic
 2012 Milan, Italy
 2014 Berlin, Germany
 2015 Gaithersburg, USA
 2016 Stockholm, Sweden
 2017 Berlin, Germany
 2018-10-08, Finland
 2020-05-28, Virtual via Zoom
 2021-05-26, Virtual via Zoom
 2022-05-19, Virtual via Zoom

Meeting, planned

2023-05-25, Stockholm, Sweden

Committee structure

ISO/TC 12 has one working group directly under the TC:

- **WG 21** “Quantities and units – General”, “*Revision of ISO 80000-1 “Quantities and units— Part 1: General”*”.
Convenor: Prof. Leslie Pendrill

Meeting held:

2020-02-27 via Zoom
 2020-05-27, via Zoom
 2021-01-21, via Zoom
 2021-02-19, via Zoom
 2021-03-19, via Zoom
 2021-04-15, via Zoom
 2021-05-11, via Zoom
 2021-09-16, via Zoom

Business plan

Current ISO/TC 12 Business plan was unanimously approved by ISO/TC 12, Draft resolution 140, 2020-04-09.
[STRATEGIC BUSINESS PLAN ISO/TC 12](#)

Programme of work and published documents

The standards produced by ISO/TC 12 are listed in Annex A.
[Work programme on ISO homepage](#)



Resolutions of the last ISO/TC 12 meeting 2022

The resolutions taken and approved at the meeting 2022 are to be found in document ISO/TC 12 [N1456](#).

***Follow up on* - Resolutions of the 19th meeting of ISO/TC 12, Quantities and units Web-meeting via Zoom 2022-05-19**

Resolution 161 (Zoom 3, 2022)

Agenda item 7, Business Plan (**Ongoing**)

ISO/TC 12 decides to confirm the Business Plan version 10, doc N 1450 as uploaded on the ISO website [iso.org](https://www.iso.org) taking into account the minor changes agreed at the meeting.

P-members approving: 9

P-members disapproving: 0

P-members abstains: 0

The decision was unanimous

Resolution 162 (Zoom 4, 2022)

Agenda item 10, Report of the liaisons (**No**)

ISO/TC 12 decides to ask the secretariat to determine by the next meeting whether there is a suitable standardisation committee on mathematics with which to establish a liaison.

P-members approving: 9

P-members disapproving: 0

P-members abstains: 0

The decision was unanimous

Resolution 163 (Zoom 5, 2022)

Agenda item 14, Items for future work (**Ongoing**)

ISO/TC 12 decides to ask Dr Knopf to explore ISO/TC 12's possible future role in producing standards for the machine readability of quantities, units, and measurement results by the next meeting.

P-members approving: 9

P-members disapproving: 0

P-members abstains: 0

The decision was unanimous



Follow up on - ISO/TC 12 Resolution 146-157 taken via Zoom meeting 2021-05-26

Resolution 151 (Zoom 7, 2021)

Agenda item 10 Study on the need for revisions in the ISO 80000 series (Done)

ISO/TC 12 decides to follow the recommendation TF1: 4:2021, Doc. N 1402:

Corrected version of ISO 80000-2, Mathematics

ISO/TC 12/TF 1 decides to recommend ISO/TC 12 to prepare a corrected version that addresses the editorial comments and blatant errors in the document, see Table 1 in N1411. This excludes the comments from Finland (N1411 slide 17) which have a more technical character and will be left for the next SR (systematic review).

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

Resolution 152 (Zoom 8, 2021)

Agenda item 10 Study on the need for revisions in the ISO 80000 series (Done)

ISO/TC 12 decides to confirm ISO 80000-3:2019, *Space and time*, and at the same time to ask ISO/CS to send the document for an earlier systematic review in October 2022.

It was noted that for part 3 *Space and time*, there were equal numbers of TF1 votes for doing a technical revision, or instead waiting for the next 5-year SR.

P-members approving:	7
P-members disapproving:	0
P-members abstains:	1 (DE)

The decision was passed by majority vote



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ANNEX A

Standard and/or project under the direct responsibility of ISO/TC 12 Secretariat (3) [International harmonized stage codes.](#)

Standard and/or proj...	● IEC/CD 80000-15 Quantities and units — Part 15: Logarithmic and related quantities
Stage	30.60
ICS	01.060
Standard and/or proj...	● IEC/CD 80000-16 Quantities and units — Part 16: Printing and writing rules
Stage	30.00
ICS	
Standard and/or proj...	● IEC/CD 80000-17 Quantities and units — Part 17: Time dependency
Stage	30.00
ICS	

[13 Published ISO standards](#)

Joint work under IEC/TC 25 lead:

IEC 80000-6:2022 *Quantities and units – Part 6: Electromagnetics* is under revision. Project leader:
Prof. Göran Grimvall



ANNEX B

P-members attending ISO/TC 12 meetings 2020-2022

Attending **Not attending**

P-member	2020 Virtual	2021 Virtual	2022 Virtual
• Austria (ASI)			
• Belarus (BELST)			
• China (SAC)			
• Czech Republic (UNMZ)			
• Denmark (DS)			
• Egypt (EOS)			
• Finland (SFS)			
• France (AFNOR)			
• Germany (DIN)			
• India (BIS)			
• Italy (UNI)			
• Japan (JISC)			
• Kuwait (KOWSMD)			
• Portugal (IPQ)			
• Russian Federation (GOST R)			
• Spain (UNE)			
• Sweden (SIS)			
• Switzerland (SNV)			
• United Kingdom (BSI)			
• United States (ANSI)			